

# INTRODUCTION TO INFORMATION MANAGEMENT

ITE104 – INFORMATION MANAGEMENT

# INTRODUCTION

- INFORMATION SYSTEM HAVE BECOME AN INDISPENSABLE PART OF TODAYS SOCIETY, PARTICULARLY IN BUSINESS. WITH THE ADVENT OF INEXPENSIVE YET POWERFUL COMPUTERS, ORGANIZATION OF ALL SIZES HAVE INSTALLED COMPUTER SYSTEM TO HELP RUN THEIR OPERATION. IT IS THEREFORE CRITICAL FOR BUSINESS PROFESSIONALS TO UNDERSTAND HOW INFORMATION SYSTEM WORK.

# INFORMATION MANAGEMENT

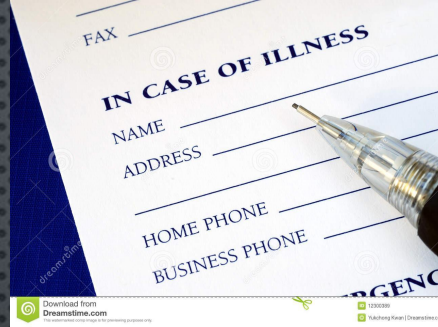
- **DATA**

- DATA IS DEFINED AS FACTS OR FIGURES, OR INFORMATION THAT'S STORED IN OR USED BY A COMPUTER. AN EXAMPLE OF DATA IS INFORMATION COLLECTED FOR A RESEARCH PAPER. THE QUANTITIES, CHARACTERS, OR SYMBOLS ON WHICH OPERATIONS ARE PERFORMED BY A COMPUTER, WHICH MAY BE STORED AND TRANSMITTED IN THE FORM OF ELECTRICAL SIGNALS AND RECORDED ON MAGNETIC, OPTICAL, OR MECHANICAL RECORDING MEDIA.

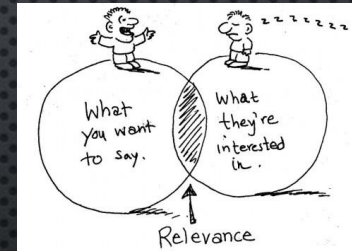
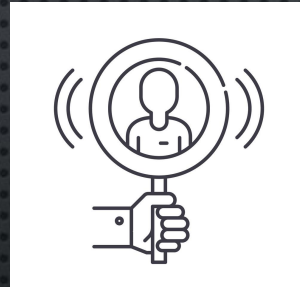
- **INFORMATION**

- THESE ARE DIFFICULT TIMES FOR ALL ORGANIZATIONS OF ALL SIZES AND IN ALL SECTORS. ON THE ONE HAND, CUSTOMERS HAVE EVER-INCREASING EXPECTATIONS IN TERMS OF THE SPEED AND QUALITY OF SERVICE THEY EXPECT AND, ON THE OTHER RESOURCES ARE CONTINUALLY UNDER PRESSURE.
- THIS DOCUMENT SETS OUT HOW EFFECTIVE INFORMATION AND RECORDS MANAGEMENT CAN HELP ANY ORGANIZATION TO MOVE FORWARD IN THIS CHALLENGING ENVIRONMENT THROUGH,
  - ACHIEVING COST AND EFFICIENCY SAVINGS;
  - MAKING BEST USE OF INFORMATION ASSETS AND
  - TAKING ADVANTAGE OF THE OPPORTUNITIES OFFERED BY NEW TECHNOLOGIES.

# CHARACTERISTICS OF INFORMATION



- **ACCURATE**
- **COMPLETE**
- **COST EFFECTIVE**
- **USER-FOCUSED**
- **RELEVANT**
- **AUTHORITATIVE**
- **TIMELY**



# SYSTEM CONCEPTS

A SYSTEM IS A GROUP OF INTERRELATED COMPONENTS WORKING TOGETHER TOWARD COMMON GOAL BY ACCEPTING INPUTS AND PRODUCING OUTPUTS IN AN ORGANIZED TRANSFORMATION PROCESS. THE CONCEPTS OF A SYSTEM ARE TECHNOLOGY, APPLICATION, DEVELOPMENT AND MANAGEMENT.

- **TECHNOLOGY**
- **APPLICATION**
- **DEVELOPMENT**
- **MANAGEMENT**



# COMPONENTS OF A SYSTEM

- THERE ARE THREE BASIC COMPONENTS OF A SYSTEM, THEY ARE
- **INPUT** – INPUT INVOLVES CAPTURING AND ASSEMBLING ELEMENTS THAT ENTER TO THE SYSTEM TO BE PROCESSED. SOME OF THE INPUTS ARE RAW MATERIALS, ENERGY, DATA ETC.
- **PROCESSING** – IT INVOLVES A TRANSFORMATION PROCESS THAT CONVERTS INPUT TO OUTPUT.
- **OUTPUT** – IT INVOLVES TRANSFORMING ELEMENT THAT HAS BEEN PRODUCED BY A TRANSFORMATION PROCESS TO THEIR ULTIMATE DESTINATION.

# TYPES OF SYSTEM

- **DYNAMIC SYSTEM** - WHEN THE INTERRELATED COMPONENT OF THE SYSTEM INTERACTS WITH EACH OTHER AND THIS CONTROLLED BY MANAGEMENT THEN IT IS KNOWN AS DYNAMIC SYSTEM.
- **CYBER NATIVE SYSTEM** - DYNAMIC SYSTEM IMPLEMENTING THE CONCEPT OF FEEDBACK AND CONTROL IS KNOWN AS CYBER NATIVE SYSTEM.
- **OPEN SYSTEM** - A SYSTEM GOT INTERACTS WITH OTHER SYSTEM IN ITS ENVIRONMENT BY EXCHANGING INPUT AND OUTPUT WITH ITS ENVIRONMENT.
- **ADOPTIVE SYSTEM** - A SYSTEM HAVING THE ABILITY TO CHANGE ITSELF AND ITS ENVIRONMENT IN ORDER TO SURVIVE IS CALLED AN ADOPTIVE SYSTEM.

# TYPES OF INFORMATION SYSTEM

- AN INFORMATION SYSTEM IS A SET OF INTERRELATED COMPONENTS THAT WORKS TOGETHER TO COLLECT, PROCESS, STORE AND BREAKDOWN THE INFORMATION TO SUPPORT DECISION MAKING.

FOLLOWING ARE THE DIMENSIONS OF INFORMATION SYSTEM:

1. **ORGANIZATION DIMENSION** - INFORMATION SYSTEM IS PART OF ORGANIZATION. INFORMATION SYSTEM WILL HAVE THE STANDARD OPERATING PROCEDURE AND CULTURE OF AN ORGANIZATION EMBEDDED WITHIN THEM. THIS INVOLVES:

- FUNCTIONAL SPECIALTIES
- BUSINESS PROCESSES
- CULTURE
- POLITICAL INTEREST GROUPS

# CONTI....

- **MANAGEMENT DIMENSION** - MANAGERS PERCEIVE BUSINESS CHALLENGES IN THE ENVIRONMENT. INFORMATION SYSTEMS SUPPLY TOOLS AND INFORMATION NEEDED BY THE MANAGERS TO ALLOCATE, COORDINATE AND MONITOR THEIR WORK, MAKE DECISION, CREATE NEW PRODUCTS AND SERVICES AND MAKE LONG RANGE STRATEGIC DECISION.
- **TECHNOLOGY DIMENSION** - MANAGEMENT USES TECHNOLOGY TO CARRY OUT THEIR FUNCTIONS. IT CONSISTS OF — COMPUTER HARDWARE/SOFTWARE, DATA MANAGEMENT TECHNOLOGY, NETWORKING/TELECOM TECHNOLOGY. IT'S ONE OF THE MANY TOOL'S MANAGERS USE TO COPE WITH THE CHANGE.
- INFORMATION SYSTEMS ARE CLASSIFIED BY ORGANIZATIONAL LEVELS, MODE OF DATA, PROCESSING, SYSTEM OBJECTIVES AND TYPE OF SUPPORT PROVIDED.

# THE FOLLOWING ARE THE TYPES OF INFORMATION SYSTEM:

- THE FOLLOWING ARE THE TYPES OF INFORMATION SYSTEM:

## 1. TRANSACTION PROCESSING SYSTEM (TPS)

- TRANSACTION PROCESSING SYSTEM ARE INFORMATION SYSTEM THAT PROCESSES DATA RESULTING FROM THE OCCURRENCES OF BUSINESS TRANSACTIONS
- THEIR OBJECTIVES ARE TO PROVIDE TRANSACTION IN ORDER TO UPDATE RECORDS AND GENERATE REPORTS TO PERFORM STORE KEEPING FUNCTION.
- THE TRANSACTION IS PERFORMED IN TWO WAYS: BATCHING PROCESSING AND ONLINE TRANSACTION PROCESSING.
- EXAMPLE: BILL SYSTEM, PAYROLL SYSTEM, STOCK CONTROL SYSTEM.

# CONTI...

## 2. MANAGEMENT INFORMATION SYSTEM (MIS)

- MANAGEMENT INFORMATION SYSTEM IS DESIGNED TO TAKE RELATIVELY RAW DATA AVAILABLE THROUGH A TRANSACTION PROCESSING SYSTEM AND CONVERT THEM INTO A SUMMARIZED AND AGGREGATED FORM FOR THE MANAGER, USUALLY IN A REPORT FORMAT. IT REPORTS TENDING TO BE USED BY MIDDLE MANAGEMENT AND OPERATIONAL SUPERVISORS.
- MANY DIFFERENT TYPES OF REPORT ARE PRODUCED IN MIS. SOME OF THE REPORTS ARE A SUMMARY REPORT, ON-DEMAND REPORT, AD-HOC REPORTS AND AN EXCEPTION REPORT.
- EXAMPLE: SALES MANAGEMENT SYSTEMS, HUMAN RESOURCE MANAGEMENT SYSTEM.

## 3. DECISION SUPPORT SYSTEM (DSS)

- DECISION SUPPORT SYSTEM IS AN INTERACTIVE INFORMATION SYSTEM THAT PROVIDES INFORMATION, MODELS AND DATA MANIPULATION TOOLS TO HELP IN MAKING THE DECISION IN A SEMI-STRUCTURED AND UNSTRUCTURED SITUATION.
- DECISION SUPPORT SYSTEM COMPRISES TOOLS AND TECHNIQUES TO HELP IN GATHERING RELEVANT INFORMATION AND ANALYZE THE OPTIONS AND ALTERNATIVES, THE END USER IS MORE INVOLVED IN CREATING DSS THAN AN MIS.
- EXAMPLE: FINANCIAL PLANNING SYSTEMS, BANK LOAN MANAGEMENT SYSTEMS.

# CONTI...

## 4. EXPERTS SYSTEM

- EXPERTS SYSTEMS INCLUDE EXPERTISE IN ORDER TO AID MANAGERS IN DIAGNOSING PROBLEMS OR IN PROBLEM-SOLVING. THESE SYSTEMS ARE BASED ON THE PRINCIPLES OF ARTIFICIAL INTELLIGENCE RESEARCH.
- EXPERTS SYSTEMS IS A KNOWLEDGE-BASED INFORMATION SYSTEM. IT USES ITS KNOWLEDGE ABOUT A SPECIFY ARE TO ACT AS AN EXPERT CONSULTANT TO USERS.
- KNOWLEDGEBASE AND SOFTWARE MODULES ARE THE COMPONENTS OF AN EXPERT SYSTEM. THESE MODULES PERFORM INFERENCE ON THE KNOWLEDGE AND OFFER ANSWERS TO A USER'S QUESTION.

# INTRODUCTION TO INFORMATION MANAGEMENT VIDEO

